

**An
Industrial Training Report**

Submitted to

UKA TARSADIA UNIVERSITY

in partial fulfillment of the requirements for the degree of

**Bachelor of Technology
in
Information Technology**

By

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Under the Guidance

of

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May 2025

CERTIFICATE

This is to certify that project work embodied in this report entitled **Industrial Training** as carried out by **Jeyansh Gandhi (202103103510289)** under my guidance in partial fulfilment of the degree of Bachelor of Technology in Information Technology, Chhotubhai Gopalbhai Patel Institute of Technology, UTU, Bardoli during the academic year 2024-2025.

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Signature of Examiner



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ERRORX

IT SOLUTION COMPANY

We Provide All kind of IT Solution.
From Mobile App to Web Development.
AI & Machine Learning Company

Date: 05-05-2025

CERTIFICATE

This is to certify that **Mr. Jeyansh Vikaskumar Gandhi** student of **Uka Tarsadiya University, Bardoli**, pursuing **Degree Information Technology** has successfully completed **6 months of internship** as a partial requirement of Short term internship in our Organization.

During this training period he done Projects on **ML & AI Development** and tasks associated etc.

His performance throughout the internship period was **excellent**, and he displayed qualities of **hardworking and co-operative besides being very punctual in attending training and er work schedule, dedication, and a willingness to learn.**

We wish **Mr. Jeyansh Vikaskumar Gandhi** all the best in his future endeavors.



Authorized Signatory,

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Company Profile

ErrorX IT Solution is a professional software development and digital transformation consultancy based in Navsari, Gujarat. Established in July 2020 by founder and full-stack developer **Vivek Patel**, the company has steadily grown into a reliable IT partner for businesses looking to digitize their operations and streamline their services. With a focus on innovation, adaptability, and technical precision, ErrorX IT Solution serves startups, SMEs, and enterprise clients across multiple industries.

ErrorX IT Solution was founded with the vision of becoming a modern, client-first software firm that emphasizes quality, innovation, and transparency. The company's mission is to deliver scalable and impactful digital solutions that empower organizations to thrive in a technology-driven world. By combining development expertise with design thinking and business insight, ErrorX aims to help clients achieve long-term digital success.

ErrorX offers a wide range of IT services tailored to the unique requirements of each client. These include **web application development**, **UI/UX design**, **custom software solutions**, and **digital marketing**. Their development stack includes modern technologies like React.js, Node.js, Python, and other open-source platforms. The company also specializes in crafting user-centric interfaces and responsive designs that enhance digital user experiences across platforms.

Led by **Vivek Patel**, who brings years of experience in full-stack development and software architecture, the team at ErrorX is composed of passionate developers, designers, and consultants. Their collaborative and agile approach ensures fast delivery without compromising quality. ErrorX emphasizes strong communication, project transparency, and post-deployment support, making them a reliable partner for long-term tech collaboration.

With a strong foundation in digital transformation and client-centric development, ErrorX IT Solution is poised for continued growth in the rapidly evolving tech landscape. The company aims to expand its services to include AI-based solutions, mobile app development, and enterprise-level consulting. They also plan to grow their presence beyond Gujarat, catering to a broader national and international clientele. Their ongoing investment in talent, tools, and training ensures they stay at the forefront of the digital age.



ACKNOWLEDGEMENT

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Jeyansh Gandhi (202103103510289)

ABSTRACT

*The **Employee Facial Attendance System** is an efficient employee attendance management system using facial recognition technology. The system is designed to streamline attendance tracking, enhance security, and simplify HR processes by automating key functions. The web application has been developed using Flask and integrates facial recognition for both admin/HR login and employee registration, ensuring secure and accurate attendance management.*

*The core features include a **Home Page**, which provides a comprehensive view of attendance data, including yearly, monthly, and weekly graphs, attendance status (present, absent, late), and a ranking of the top 10 employees with the highest attendance. The **Employee List Page** allows HR to search for employee profiles using a name-based search bar, with the ability to view the complete list of employees.*

*The **Employee Registration Page** enables HR to add new employees by inputting personal details, uploading necessary documents (e.g., resume, certificates), and performing face detection to register the employee's face for future attendance tracking. The **Project Completed Page** provides an overview of completed projects, including company names, project types, start and completion dates, and budgets. The **Ongoing Projects Page** displays similar information for projects that are still in progress, with updated timelines and completion statuses.*

*The system also includes a **Mailing Option**, allowing HR to send common instructions to all employees via a form with subject, CC, and body fields. Additionally, the **Excel Export Button** allows HR to download a monthly attendance sheet in Excel format for record-keeping.*

Finally, the login process for both HR and admin is secured using facial recognition technology, ensuring that only authorized personnel can access sensitive employee data. This project aims to provide a user-friendly, secure, and automated solution for employee attendance management.

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CHAPTER 1 INTRODUCTION

1.1 Overview

The **Employee Facial Attendance System** is an innovative and automated solution designed to streamline the attendance management process within organizations. This system utilizes facial recognition technology to mark attendance for employees, reducing the chances of errors and fraud often associated with traditional methods like manual entries or fingerprint-based systems. The application aims to improve security, simplify HR operations, and ensure more accurate attendance tracking.

The system integrates various modules including employee registration, project management, attendance tracking, and an admin panel for HR to manage data effectively. By leveraging facial recognition for both admin login and employee attendance marking, the system enhances accessibility and reduces human intervention, ensuring better data integrity.

1.2 Problem Definition

Traditional attendance management systems are often prone to errors, manipulation, and inefficiency. Manual methods like signing registers or punching cards may lead to inaccurate data entry, tardiness, or even fraudulent attendance marking (e.g., buddy punching). Additionally, managing employee records, project details, and communications in a centralized and effective way can become cumbersome for HR departments.

The problem addressed by this project is the need for a more secure, accurate, and automated attendance system that minimizes human error and fraud. Moreover, organizations require a solution that not only tracks attendance but also integrates project management and communication functionalities for streamlined HR operations.

1.3 Scope/Application

The **Employee Facial Attendance System** can be implemented across various industries and organizations to enhance the efficiency of attendance tracking and management. The scope of this project includes the following applications:

1. **Corporate Offices and Businesses:** This system can replace traditional attendance methods like manual registers or biometric machines, providing a more secure, efficient, and automated solution using facial recognition. It ensures accurate attendance tracking with minimal human intervention.
2. **Educational Institutions:** Schools, colleges, and universities can adopt this system to track the attendance of both faculty and students. It eliminates the need for manual entry, reduces the chances of errors, and makes attendance management faster and more reliable.
3. **Manufacturing and Industrial Sectors:** In factories or industrial settings, where shift-based work is common, this system can automate attendance tracking, ensuring that employees are marked present on time, thus improving operational efficiency and reducing manual errors.
4. **Healthcare Facilities:** Hospitals, clinics, and other healthcare institutions can use this system to monitor staff attendance accurately. It helps in managing shift rotations, ensuring timely attendance of healthcare professionals, and maintaining compliance in critical environments.

CHAPTER 2 TRAINING ACTIVITIES

Week 1 & 2: Basic Image Processing and Transformation

Over the past 15 days of my internship, I explored OpenCV and worked on various image processing tasks. I started with basic operations such as image resizing, saving, and color conversion, using PIL for opening and handling color transformation. I then applied different image filtering techniques, including blur, contour, emboss, and edge enhancement. These filtering techniques allowed me to improve the quality of the images and make them more suitable for further analysis.

Additionally, I performed image-to-array conversion and modified color using arrays. I also generated images directly from arrays and experimented with different ways to manipulate them. This process gave me a better understanding of how images are processed at the pixel level. Furthermore, I inserted various shapes into images, exploring how to overlay geometric elements into existing images. These tasks helped me to understand the impact of different transformations on the final output.

Week 3 & 4: Advanced Filtering and Image Manipulation

In these weeks, I continued to work with OpenCV and focused on advanced image manipulation. I experimented with various filtering techniques, such as thresholding and Canny edge detection, to identify edges and important features within images. These filters allowed me to detect contours, which helped in analyzing shapes and boundaries of objects in the image.

Additionally, I applied Pyramid Contours and Convex Hull techniques to study the structure of objects in images. These techniques are especially helpful in recognizing the shape of objects and understanding their spatial relationship. I also explored momentum-based methods to track motion in video, which enhanced my ability to detect moving objects in dynamic scenes. The use of hand and object detection helped me track specific objects in real-time.

During this period, I also worked on image and video background removal, utilizing advanced algorithms like GrabCut to isolate foreground objects from their backgrounds. This technique is vital for segmentation and object tracking, especially in real-time applications. These tasks helped me improve my ability to manipulate and extract useful information from images.

Week 5 & 6: Advanced Detection Techniques

In these weeks, I gained hands-on experience with several advanced detection techniques. I explored Hough Circle Detection, a method for identifying circular objects in images, which is commonly used in applications such as detecting wheels, coins, or traffic signs. I also worked on corner detection methods, allowing me to identify key feature points in images, which is essential for image matching and recognition.

Moreover, I explored video background removal, which is essential for isolating objects in dynamic scenes. This technique helped me understand the importance of object segmentation in videos and how to handle moving objects. I applied these concepts in real-world applications by experimenting with Optical Flow using OpenCV. This technique allows the analysis of motion between frames, which is crucial for tracking objects across video streams.

Week 7 & 8: Object Tracking, Detection, and Real-Time Applications

In these weeks, I continued to enhance my understanding of object tracking and detection. I focused on Optical Flow, which allowed me to track object movement across video frames, an essential aspect of real-time applications. Additionally, I worked on object tracking techniques, which are crucial for real-time video analysis, and applied them to detect and track specific objects in videos.

I also worked on template matching, which involves detecting specific objects within a larger image or video frame. This method is commonly used for object detection in images and real-time video feeds. I further applied background removal techniques in videos to focus on moving objects, enhancing my skills in segmentation and motion analysis.

To apply the concepts practically, I worked on a project that involved classifying dogs and cats using object tracking and background removal techniques. This hands-on experience allowed me to refine my skills in real-time applications, motion detection, and object classification.

CHAPTER 3 SYSTEM PLANNING

3.1 Project Development Approach

- Iterative Model
- The **Iterative Model** is a software development process where the project is divided into smaller parts or modules. Each module is developed through repeated cycles (iterations), and feedback is gathered after each cycle to improve the system. This model focuses on creating an initial version of the software quickly and then refining it in successive iterations, with each iteration enhancing or expanding the system's capabilities. At the end of each iteration, a prototype or a working version of the software is delivered for testing, and changes are made based on user feedback. This approach allows for gradual improvement and ensures that the system is developed in manageable phases.
- The **Iterative Model** was chosen for the **Employee Facial Attendance System** because it provides flexibility and allows for continuous refinement throughout the development process. Given the complexity of the project, which involves various modules such as facial recognition, employee registration, attendance tracking, and admin management, the iterative approach ensures that each module can be developed, tested, and improved independently. This iterative approach also facilitates the incorporation of new requirements or improvements based on feedback gathered after each cycle, allowing the system to evolve according to changing needs. Additionally, the iterative model supports early testing of features, helping identify and address issues promptly, leading to a more reliable and efficient system.
- **Improved Quality:** Frequent testing and feedback after each iteration result in higher-quality software, with issues being addressed early.
- **Manageable Phases:** Breaking the project into smaller iterations makes the development process more manageable, reducing the risk of overwhelming complexity.
- **Adaptability to Changes:** Changes and enhancements can be easily incorporated into the project after each iteration, ensuring that the software meets evolving needs.

- **Faster Delivery of Features:** By delivering a working version after each iteration, users and stakeholders can begin testing and using features sooner, allowing for quicker deployment.
- **Reduced Risk:** Continuous evaluation during each iteration allows the development team to spot risks and address them early in the project lifecycle.

3.2 System Modules

The **Employee Facial Attendance System** consists of several key modules that work together to provide a comprehensive and efficient employee attendance management system. Below are the primary modules of the system.

3.2.1 Admin/HR Login and Registration (Facial Recognition)

This module allows HR or the admin to log in to the system using facial recognition for authentication. New users (HR/Admin) can also register their facial data for secure login. It ensures secure access to the system and allows HR/Admin to manage the attendance system and employee data efficiently.

3.2.2 Home Page

The home page provides an overview of attendance statistics, such as total present, absent employees, and top performers based on attendance. It serves as the dashboard for HR/Admin to quickly assess overall attendance performance, with visual graphs to help make informed decisions.

3.2.3 Employee List Page

This page lists all employees along with their attendance status. A search bar is available to find employees by name. It helps HR/Admin search for individual employee profiles and track their attendance history, making employee management easy.

3.2.4 Employee Registration Page

HR/Admin can register new employees here by inputting personal details and uploading necessary documents. Additionally, the employee's face is captured and stored for future attendance tracking. It facilitates the onboarding of new employees into the system, ensuring that their personal details and facial data are properly stored for future recognition.

3.2.5 Completed Projects Page

This page displays details about the projects completed by employees, such as project names, dates, budget, and company names. It helps track and manage the status of projects that have been finished, allowing HR/Admin to monitor completed tasks and project performance.

3.2.6 Ongoing Projects Page

Lists projects that are currently ongoing, providing details about their status, start dates, and estimated completion times. This module helps HR/Admin keep track of ongoing tasks and deadlines, ensuring timely completion and resource management.

3.2.7 Mailing Option

A mail form allows HR/Admin to send bulk messages or announcements to all employees. It includes fields for subject, CC, and body of the email. It streamlines communication by enabling HR/Admin to quickly send messages or instructions to all employees at once.

3.2.8 Attendance Excel Export

This module allows HR/Admin to export the monthly attendance data into an Excel sheet, which can be downloaded for analysis or record-keeping. It provides a convenient way to back up attendance records or analyze data offline, ensuring efficient record management.

3.2.9 Logout

This module logs out the current user, ensuring the security of the system by ending the session. It ensures system security by allowing users to log out securely, preventing unauthorized access to the system after use.

3.3 Functional Requirements

Table 3.3 Functional Requirements

ID	Title and Description
FR1	Title: Facial Recognition for Login and Registration: Desc: The system shall allow HR/Admin to log in and register employees using facial recognition. During registration, the system shall capture and store the employee's face data for future login and attendance recognition.
FR2	Title: Employee Profile Management Desc: The system shall allow HR/Admin to view, update, and search employee profiles by name or employee ID. The system shall store and display personal details, attendance records, and relevant documents (e.g., resumes, certificates) in each employee's profile.
FR3	Title: Attendance Tracking and Marking Desc: The system shall automatically mark attendance for employees based on facial recognition when they log in to the system or when they enter the company premises (via a camera). The system shall display attendance status (present, absent, late) for each employee for a given period (daily, weekly, monthly).
FR4	Title: Attendance Reporting and Exporting Desc: The system shall generate attendance reports for HR/Admin, displaying a summary of employee attendance, including presence, absence, and lateness. The system shall allow HR/Admin to export attendance data into an Excel sheet for record-keeping and analysis.
FR5	Title: Project Tracking and Management Desc: The system shall allow HR/Admin to view completed and ongoing projects, with details such as company name, project type, timeline, and status. The system shall allow HR/Admin to manage project information for record-

	keeping.
FR6	Title: Employee Registration for New Employees Desc: The system shall allow HR to add new employees by entering personal information and capturing their face data for future identification and attendance. The system shall allow HR to upload employee-related documents (e.g., certificates, resumes).
FR7	Title: Secure User Authentication Desc: The system shall ensure secure user authentication using facial recognition for HR/Admin logins. The system shall maintain secure sessions and prevent unauthorized access.
FR8	Title: System Logout Desc: The system shall provide the option to log out securely to prevent unauthorized access after the HR/Admin has completed their tasks.

3.4 Non Functional Requirements

3.4.1 Performance

The system should process facial recognition and attendance marking within 2–3 seconds for smooth user experience. The application should support real-time video feed processing for accurate face detection.

3.4.2 Scalability

The system should be scalable to handle increasing numbers of employees without performance degradation. It should support future integration with more advanced modules or additional branches of the company.

3.4.3 Security

The system must securely store facial data and employee information using proper encryption techniques. Only authorized HR/Admin personnel should have access to sensitive data and system configuration.

3.4.4 Usability

The system should have a user-friendly interface, allowing HR/Admin to easily navigate and manage attendance, profiles, and reports. All forms and buttons should be intuitive and clearly labeled. Reliability The system should maintain consistent performance without crashing or losing data. It should handle unexpected situations (e.g., poor lighting or failed face detection) gracefully.

3.4.5 Maintainability

The codebase should be modular and easy to update or debug. Future modifications such as adding new features should be easily implementable.

3.4.6 Availability The system should be available during working hours (e.g., 9 AM to 6 PM) without downtime. In case of any failure, it should have quick recovery support.

3.5 Timeline Chart

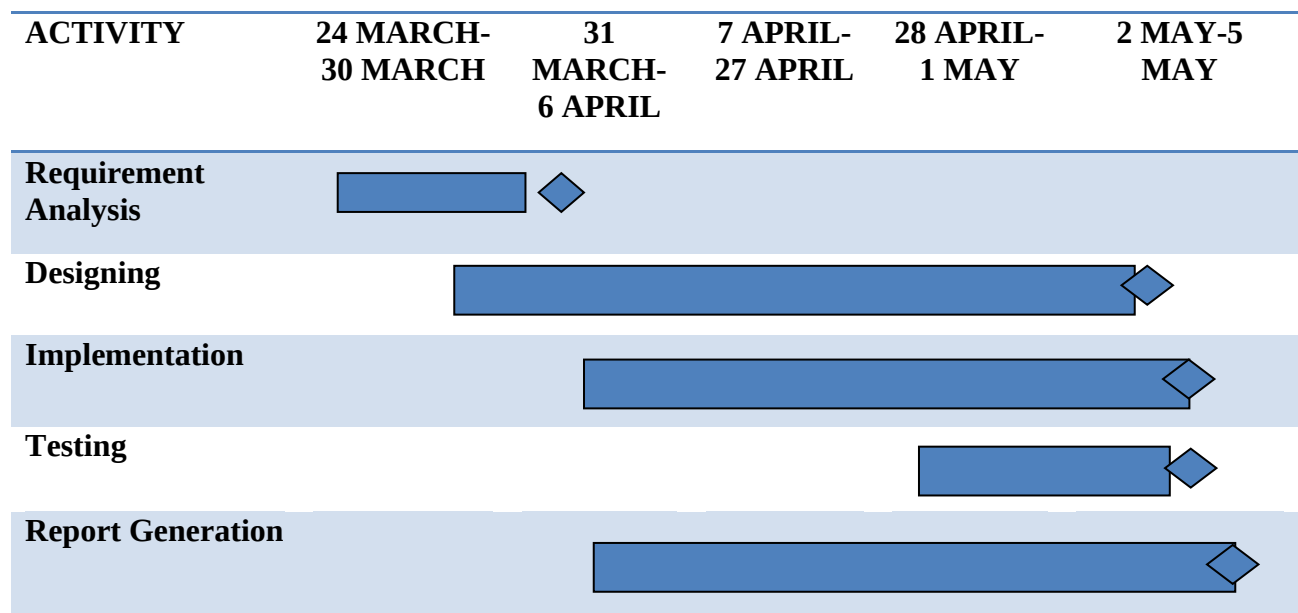


Figure 3.1 Sample Timeline Chart

CHAPTER 4 SYSTEM DESIGN

4.1 Use Case Diagram

The use case diagram shows the interaction between the **Admin** (the only user) and the system. It highlights key functionalities like **admin login (via face recognition)**, **registering employees**, **marking attendance via CCTV**, **viewing employee data**, **managing projects**, and **sending group emails**. It provides a clear overview of what the system allows the admin to do.

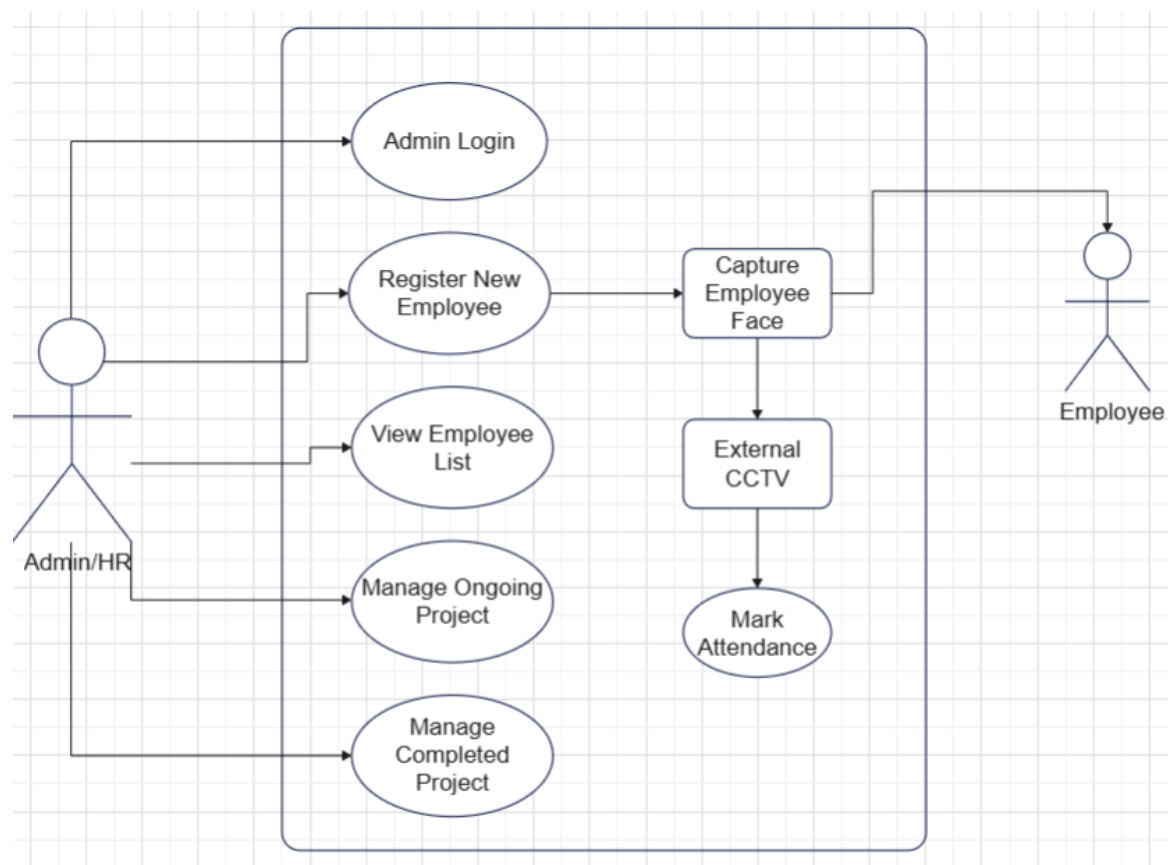


Figure 4.1: Use Case Diagram

4.2 Sequence Diagram

The sequence diagram represents the step-by-step flow of how the admin interacts with the system. For example, in the “Register Employee” process, it shows how data is entered, the face is captured, and everything is stored in the database. It focuses on the **order of operations between objects** like the Admin, UI, Face Detection Module, and Database.

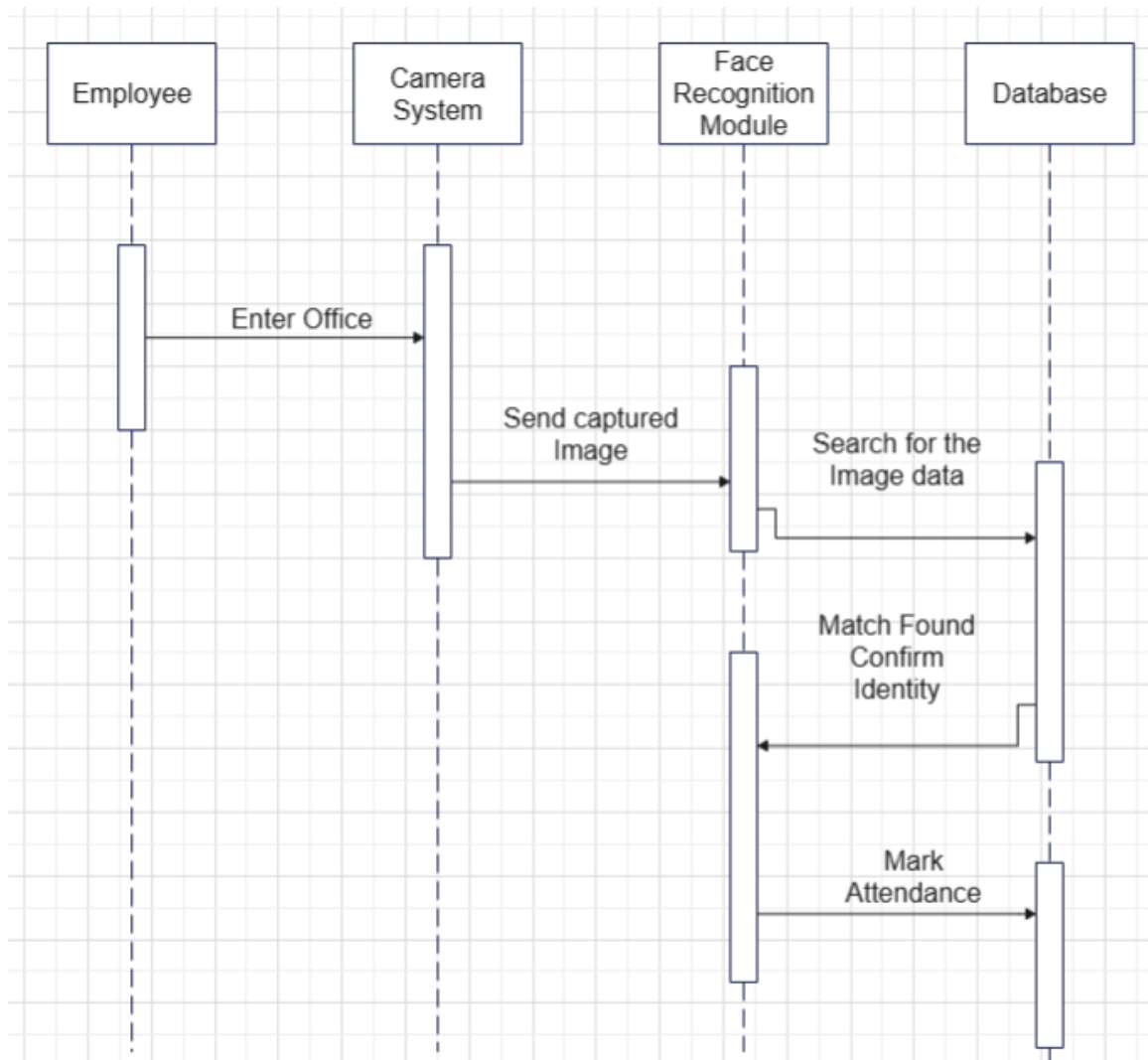


Figure 4.2: Sequence Diagram

4.3 Activity Diagram

The activity diagram illustrates the flow of actions in the system like a flowchart. It shows the decision-making steps and processes such as **admin login**, **employee registration**, and **marking attendance**. This helps to understand the logical flow of activities from start to end for each core feature.

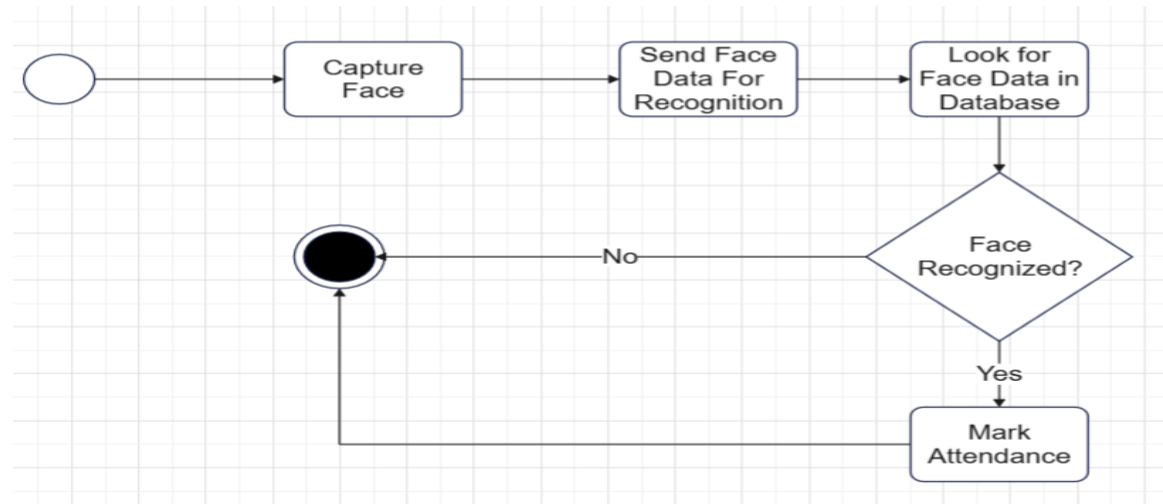


Figure 4.3: Activity Diagram

4.4 Class Diagram

The class diagram represents the **structure of the system** with classes like Admin, Employee, FaceRecognition, Attendance, Project, and MailService. It shows class attributes, methods, and how they are related (1-to-many or dependent). This helps in understanding the **backend logic and relationships** between components.

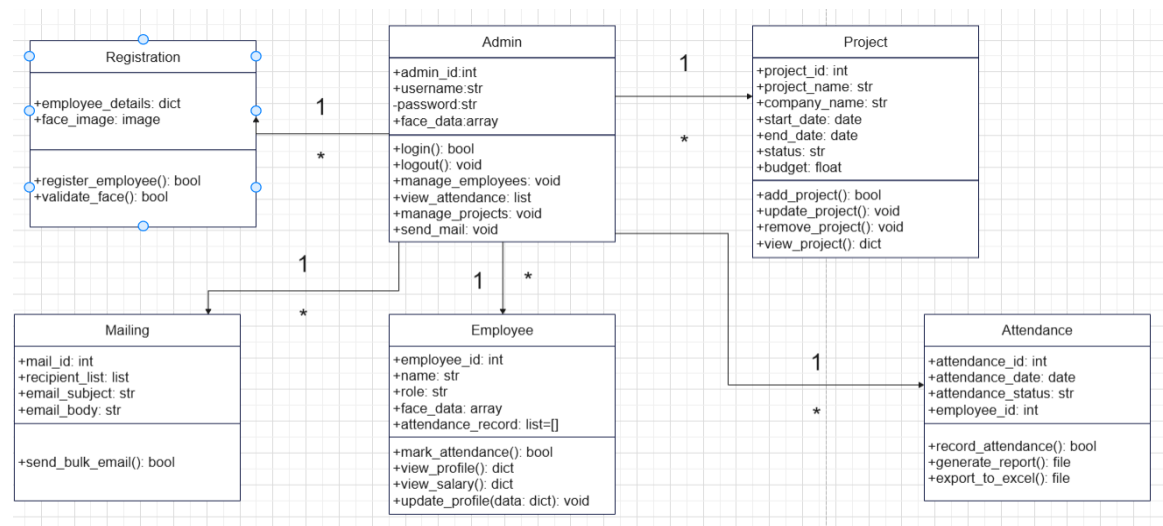


Figure 4.4: Class Diagram

4.5 Database Schema

Table 4.1 Employees Table

Column Name	Data Type	Size	Constraint	Description
Employee_name	VARCHAR	5	---	Name of Employee
Employee_id	VARCHAR	50	Primary Key	Employees particular id
Email	VARCHAR	50	---	Official email of employee
Phone	NUMBER	10	---	Employees phone number
Address	VARCHAR	50	---	Address of the employee
Position	VARCHAR	50	---	Position at which employee is working
Department	VARCHAR	50	---	In which department he/she is working
DOB	DATE	---	---	Date of the employee
DOJ	DATE	---	---	Date when he/she joined

				the firm
Resume	DOCUMENT	---	---	Resume of the employee
Offer_Letter	DOCUMENT	---	---	Offer letter provided to the employee
Bond_Document	DOCUMENT	---	---	Contract document
Captured_Image_Path	IMAGE	---	---	Image of the employee
Embeddings	VARCHAR	---	---	Decrypted vectors of employees face feature
Key	VARCHAR	---	---	Key to encrypt the embeddings

Table 4.2 Attendance Table

Column Name	Data Type	Size	Constraint	Description
Employee_name	VARCHAR	5	---	Name of Employee
Time	DATE	50	---	Time of the attendance marked
Attendance	VARCHAR	50	---	Present or

				Absent
--	--	--	--	--------

Table 4.3 Project Table

Column Name	Data Type	Size	Constraint	Description
Project_name	VARCHAR	5	---	Name of Project
Company_name	VARCHAR	50	Primary Key	Name of the company
Start_date	DATE	50	---	Project starting date
End_date	DATE	10	---	Project ending date
Budget	NUMBER	50	---	Total budget of the project
Project_type	VARCHAR	50	---	Which technology is used in the project
Description	VARCHAR	50	---	Description of the project
Status	VARCHAR	---	---	Completed or Ongoing

4.6 Data Flow Diagram

The DFD shows how data flows through the system.

Level 0: High-level interaction between Admin, System, and Database.

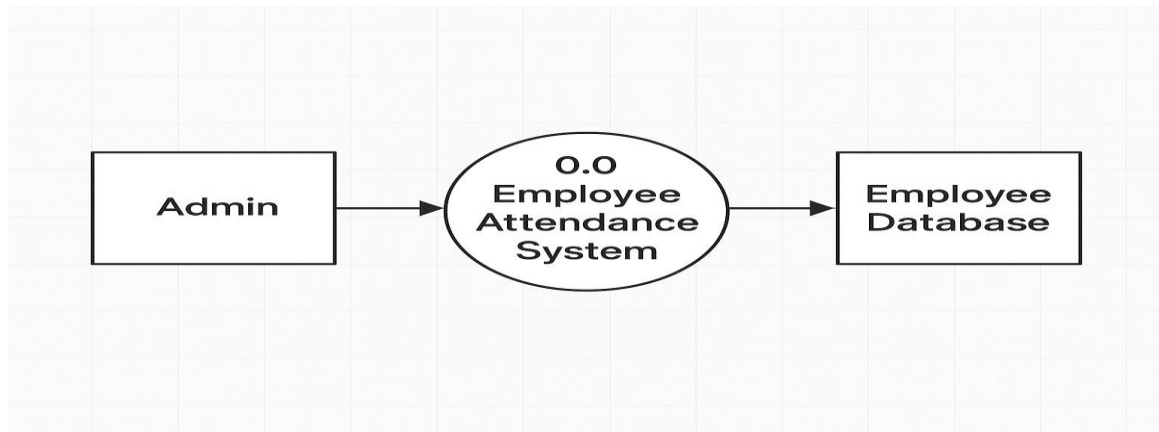


Figure 4.6.1: Level 0 Data Flow Diagram

Level 1: Breaks the system into processes like register, mark attendance, etc.

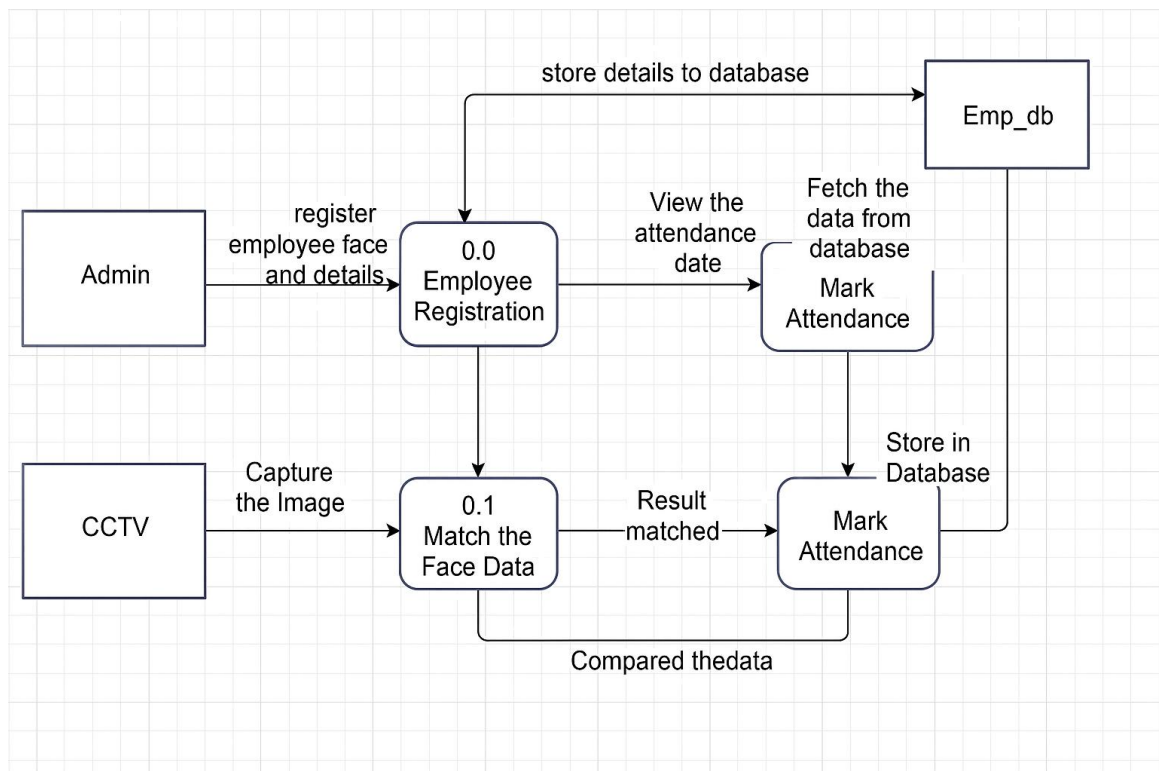


Figure 4.6.2: Level 1 Data Flow Diagram

Level 2: Provides detailed logic of core processes like face matching and saving attendance.

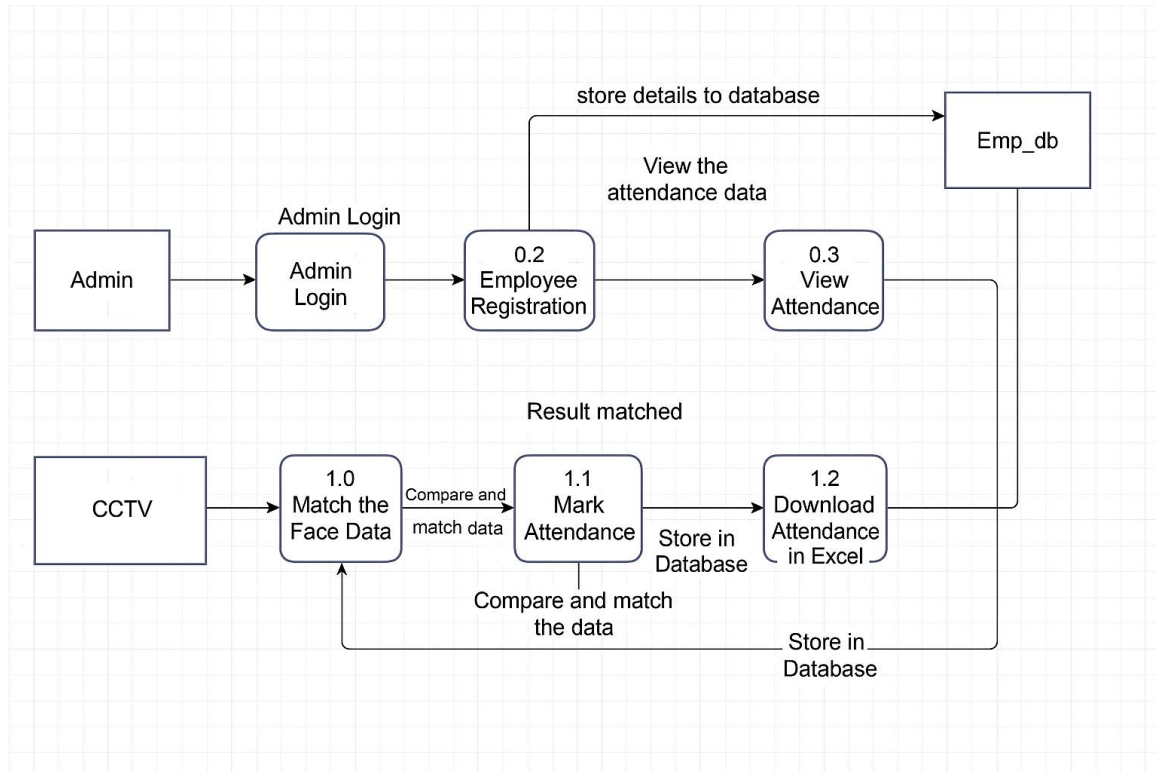


Figure 4.6.3: Level 2 Data Flow Diagram

CHAPTER 5 IMPLEMENTATION AND TESTING

5.1 Hardware and Software Requirements

5.1.1 Hardware Requirements

- **Processor:** Intel i5 or higher
- **RAM:** 8GB or more
- **Storage:** 256GB SSD or higher
- **Camera:** Integrated webcam or external CCTV/IP camera
- **Monitor:** Full HD resolution (1920x1080) or better
- **Peripherals:** Keyboard and mouse
- **Internet:** Stable broadband connection for real-time data sync and updates

5.1.2 Software Requirements

- **Operating System:** Windows 10/11, Linux (Ubuntu), or macOS
- **Programming Language:** Python 3.8 or higher
- **Libraries/Frameworks:** OpenCV, NumPy, Pandas, FaceNet, dlib, Flask, Matplotlib
- **Database:** MongoDB (for storing face encodings and employee data)
- **Web Framework:** Flask (for backend development)
- **Frontend Tools:** HTML, CSS, JavaScript, Bootstrap/Tailwind CSS
- **IDE:** Visual Studio Code, Jupyter Notebook, or PyCharm
- **Version Control:** Git and GitHub (for source code management)
- **Security:** bcrypt or similar library for secure login and encryption

5.2 Snapshots

➤ Login Page

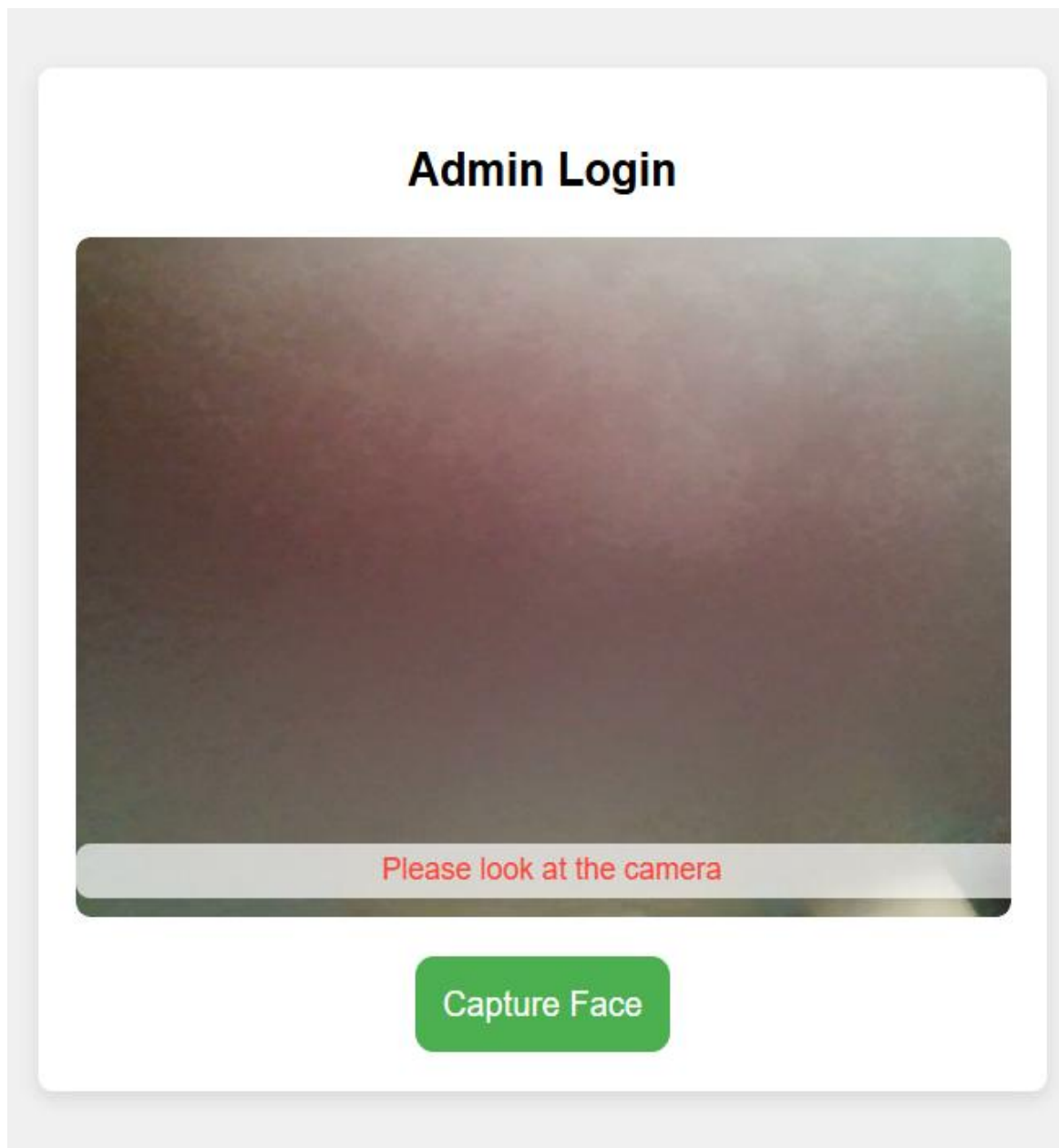


Figure 5.1 Login Page

Only authorized HR or Admin personnel can log in using facial recognition, ensuring secure and contactless access to the system.

➤ Home Page

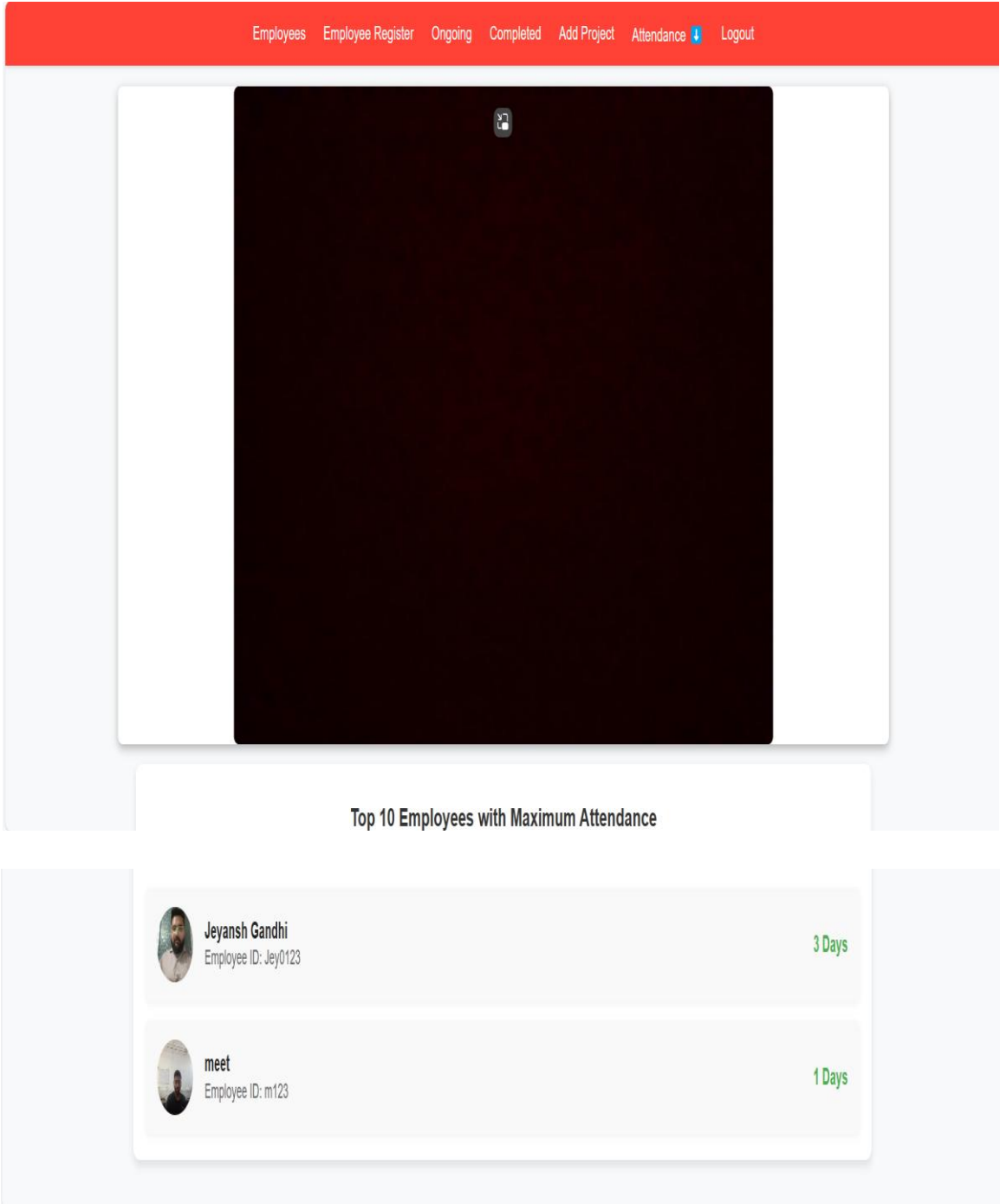


Figure 5.2 Home Page

The home page displays real-time video feed for facial recognition, along with key attendance statistics and a list of the top 10 employees with the highest attendance, providing HR/Admin a clear and interactive summary.

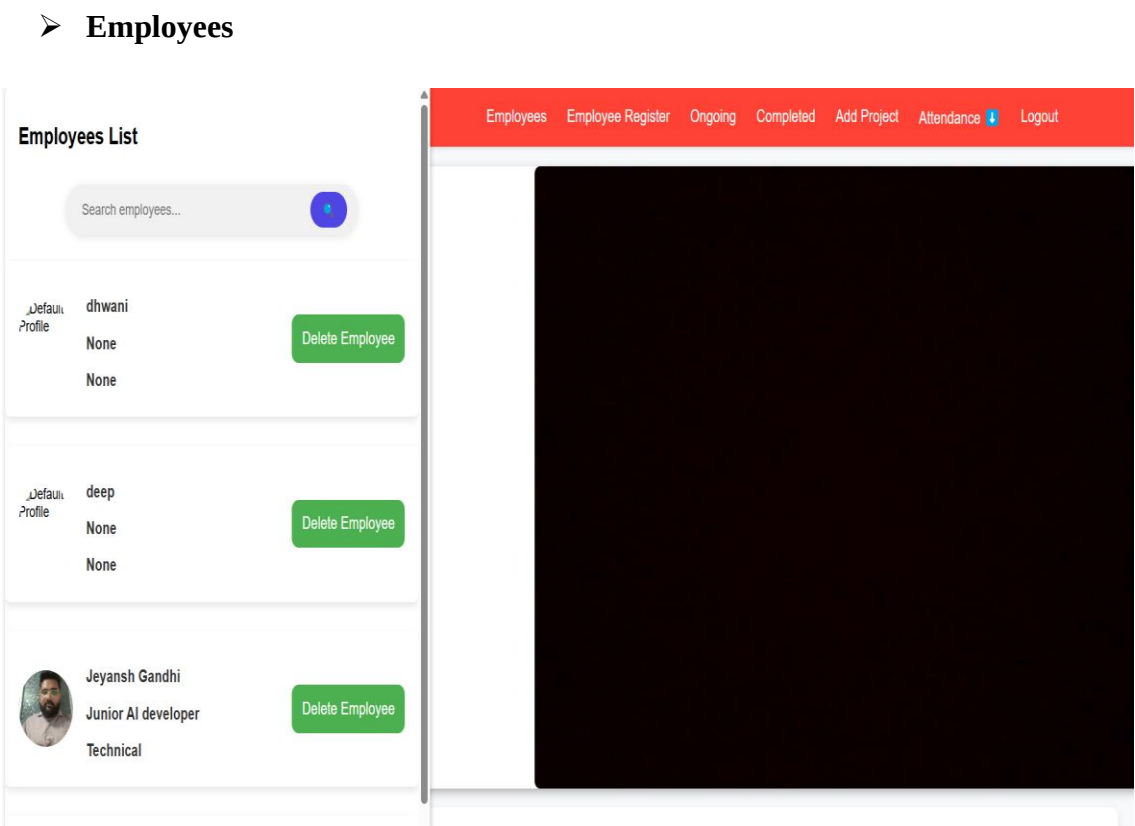


Figure 5.3 Employees

This section displays all registered employees with their photo, name, position, and department. A sidebar with a search bar allows HR/Admin to quickly find and manage employees. Each employee card includes a delete option for easy data management.

➤ **Employee Registration**

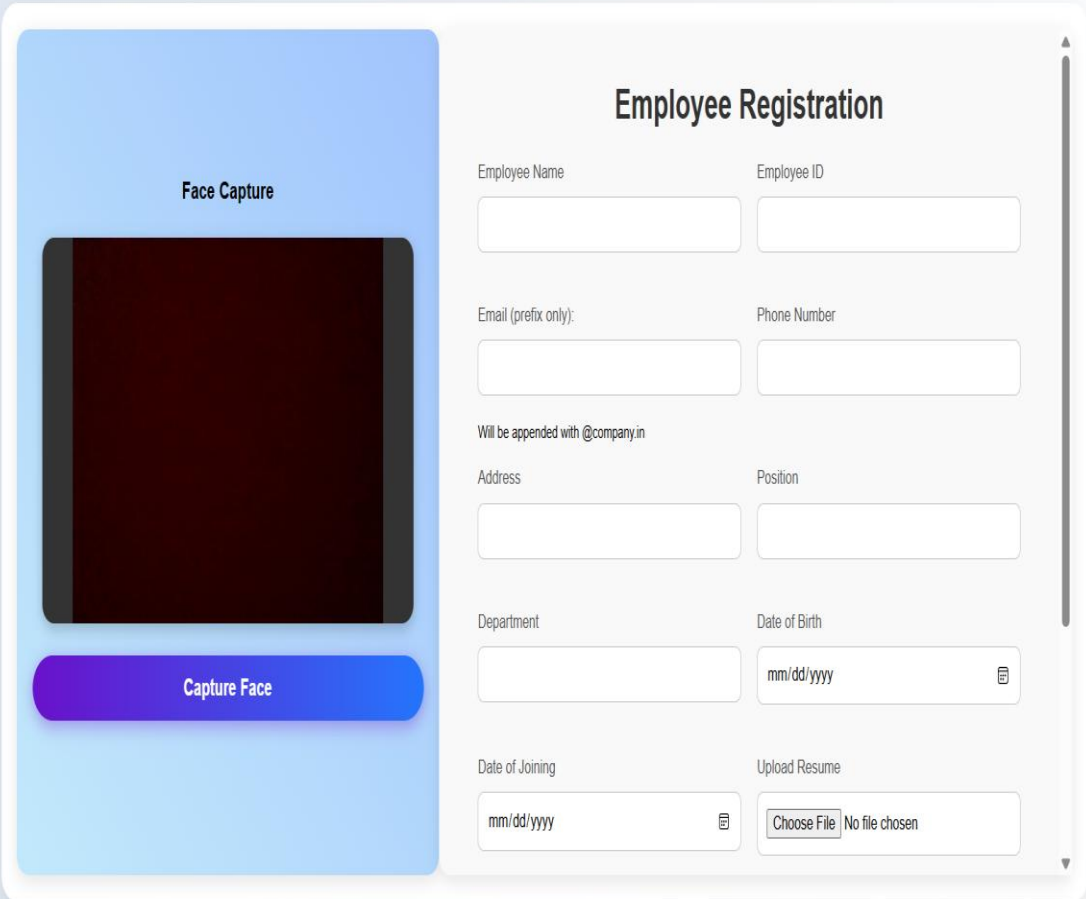
The image shows a web-based 'Employee Registration' form. On the left, there is a 'Face Capture' section with a dark rectangular area for the face and a blue 'Capture Face' button below it. The right side of the form is titled 'Employee Registration' and contains several input fields: 'Employee Name' and 'Employee ID' at the top; 'Email (prefix only):' and 'Phone Number' below them; a note 'Will be appended with @company.in' under the email field; 'Address' and 'Position' in the middle; 'Department' and 'Date of Birth' (with a date picker icon) below that; and 'Date of Joining' (with a date picker icon) and 'Upload Resume' (with a 'Choose File' button and 'No file chosen' text) at the bottom. The entire form is set against a light blue background with a subtle shadow.

Figure 5.4 Employee Registration

The registration page allows HR/Admin to add new employees by entering personal details, uploading a resume and certificates, and capturing the employee’s facial image. This face data is stored for future attendance recognition, ensuring seamless integration into the system.

➤ Ongoing Project

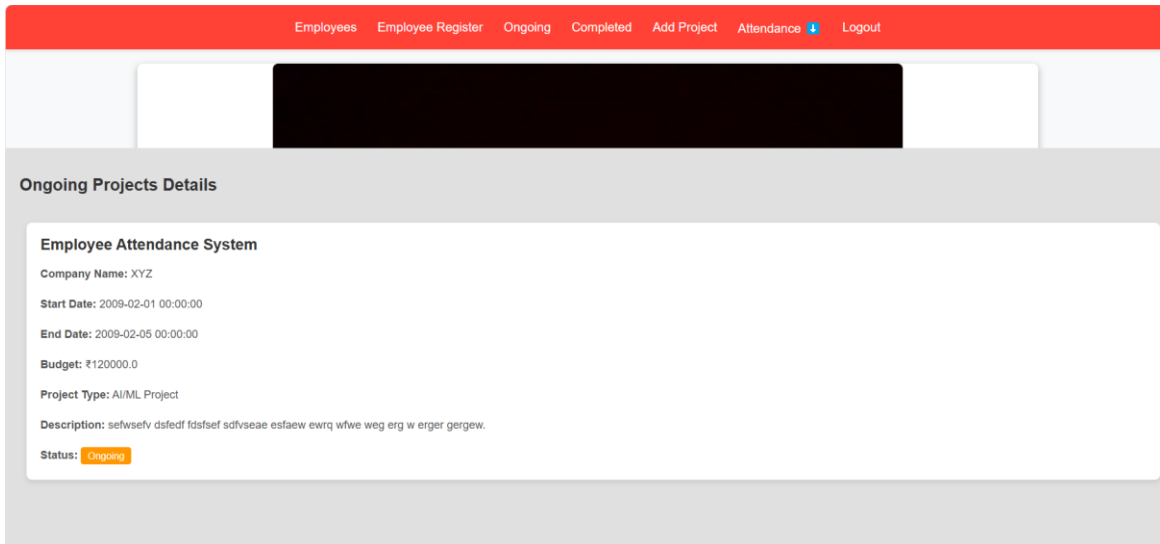


Figure 5.5 Ongoing Project

Displays a list of projects currently in progress, including project name, company, start date, estimated completion date, and project type, helping HR/Admin track active work efficiently.

➤ Completed Project

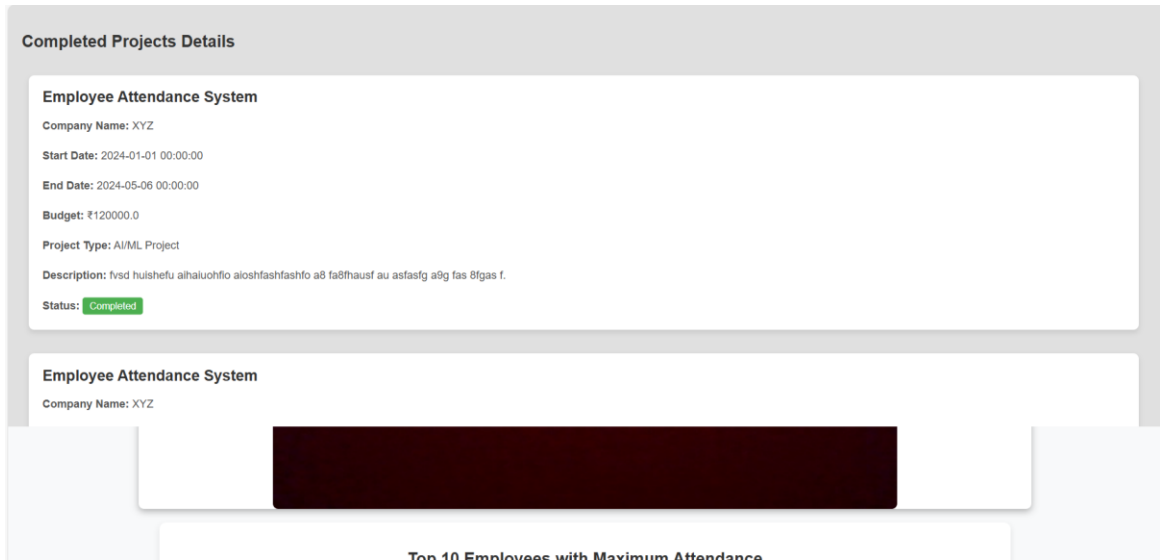


Figure 5.6 Completed Project

Displays a list of projects currently in progress, including project name, company, start date, estimated completion date, and project type, helping HR/Admin track active work efficiently.

➤ Add Project

Add New Project

Project Name

Enter project name

Company Name

Enter company name

Start Date

mm/dd/yyyy

End Date

mm/dd/yyyy

Budget (₹)

Enter budget

Project Type

Select type

Project Description

Enter project details...

Status

☐ Mark as Completed

Save Project

Figure 5.7 Add Project

This feature allows HR/Admin to add new projects by entering details like project title, company name, project type (ongoing or completed), start and end dates, estimated budget, and current status. It helps maintain a structured project database linked to employee performance and workload.

➤ Attendance

EmployeesEmployee RegisterOngoingCompletedAdd ProjectAttendance

This WeekThis MonthThis Year

Downloadsattendance_week_20250502.xlsxOpen fileSee more

Figure 5.8 Attendance

Displays a list of projects currently in progress, including project name, company, start date, estimated completion date, and project type, helping HR/Admin track active work efficiently.

5.3 Test Cases

Sample test cases are given as below in :

Table 5.1 Test Cases

TestID	Case	Test Data	Expected Result	Actual Result	Pass/Fail
1	Login	- Internet Connectivity - Face	If Face is not registered as with HR/Admin as position will give error	If Face is not registered as with HR/Admin as position will give error	Pass
2	Home	Video Feed for Recognition and top employees with attendance list	Recognize the face and display list of top attendance employees	Recognize the face and display list of top attendance employees	Pass
3	Employees	Sidebar will slide along with searchbar and profiles	Displays the profile and search bar helps to get profile	Displays the profile and search bar helps to get profile	Pass
4	Employee Registration	Register the new employees with details and face id	Should store the details in the database and shows the profile in sidebar	Should store the details in the database and shows the profile in sidebar	Pass
5	Ongoing Project	Slide up the details of the project on	A slider with heading of Ongoing Project	A slider with heading of Ongoing	Pass

		going	come from the bottom of the screen and disappears when click on screen	Prkoject come from the bottom of the screen and disappears when click on screen	
6	Completed Project	Slide down the details of the project completed	A slider with heading of Ongoing Prkoject come from the top of the screen and disappears when click on screen	A slider with heading of Ongoing Prkoject come from the top of the screen and disappears when click on screen	Pass
7	Add Project Details	Add the details of the projects	Page with input section will appear and show be saved when click saved	Page with input section will appear and show be saved when click saved	Pass
8	Attendance	Save the Attendance	It should give the list of employees present and download an excel file	It should give the list of employees present and download an excel file	Pass
9	Logout	Logout	Move to Login Page	Move to Login Page	Pass

CHAPTER 6 CONCLUSION AND FUTURE SCOPE

The *Employee Facial Attendance System* has the potential for numerous enhancements that can significantly boost its utility and adaptability. One key improvement is the addition of **real-time alerts** for absenteeism, late arrivals, or early departures, which can help HR departments take timely action and improve discipline. Integrating the system with **payroll software** would streamline salary calculations based on attendance data, making the process more accurate and efficient. Additionally, a **mobile application** can be introduced for HR and admin staff to monitor attendance, manage profiles, and communicate with employees remotely. The system can also be extended for **multi-branch connectivity**, allowing centralized attendance tracking and management across different office locations from a single dashboard.

From a technical perspective, the system can adopt more **advanced deep learning-based facial recognition models** such as ArcFace or RetinaFace, improving accuracy in conditions like poor lighting, crowded environments, or partial face visibility. To enhance security, **multi-factor authentication (MFA)** can be implemented by combining facial recognition with other methods such as voice recognition, fingerprint scanning, or OTP-based login, particularly for sensitive administrative access. Another essential feature for future updates is the ability to **register new admins using facial recognition** and allow **existing admins to update their profiles**, such as changing contact information or re-scanning facial data when needed. These enhancements would make the system more robust, secure, and scalable for real-world enterprise use.

The *Employee Facial Attendance System* successfully achieves its objective of creating a smart, contactless, and automated solution for attendance management using facial recognition technology. By eliminating manual errors, buddy punching, and time fraud, the system ensures greater transparency and reliability in tracking employee presence. It combines real-time face detection, secure data storage, and an intuitive admin interface to deliver a solution that is not only functional but also easy to maintain. Designed with scalability in mind, it can accommodate growing organizational needs and serve as a foundation for future innovations. Overall, the project stands as a modern, efficient, and intelligent approach to attendance tracking that benefits both HR teams and organizational productivity.

REFERENCES

Web references

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